Bihar Engineering University, Patna

B.Tech 4th Semester Exam., 2018

Subject: Discrete Mathematical Structure and Graph Theory

Code: 211405

Time: 03 Hours

Full Marks: 70

Instructions:-

(i) *The marks are indicated in the right-hand margin.*
(ii) *There are NINE questions in this paper.*
(iii) *Attempt FIVE questions in all.*
(iv) *Question No. 1 is compulsory.*

Q.1 Choose the correct answer (any seven): **[2x7=14]**

(a) For any three sets A, B and C, which of the following statements is wrong?

(i) Acup(*B*cupC) = (Acup*B*)capC
(ii) Acup(*B*cupC) = Acup(*B*cupC)
(iii) Acap(*B*cupC) = (Acap*B*)cup(AcapC)
(iv) None of the above

(b) Let A and B be two non-empty sets. Then the set of all ordered pairs (a, b), where *a*inA and *b*in*B* is called

(i) product set
(ii) poset
(iii) binary set
(iv) None of the above

(c) If (*a*,*a*)in*R* or equivalently *a R a*, forall*a*inA, then a relation R on a set A is called

(i) equivalent
(ii) reflexive
(iii) symmetric
(iv) anti-symmetric

(d) Let A and B be finite sets with |A| = *n* and |*B*| = m. How many functions are possible from A to B with A as the domain?

(i) n
(ii) m^m
(iii) m
(iv) m^n

(e) For the functions f and g defined by $f(x) = x^3$ and $g(x) = x^2 + 1 forall x in R$, the value of $(g \circ f)(x)$ is

(i) $x^2 + 1$
(ii) $x^3 + 1$
(iii) $x^6 + 1$
(iv) $x^5 + 1$

(f) A group G is said to be Abelian (or commutative) if for every

(i) *a*, *b*in*G*
(ii) $a \cdot b = b \cdot a$
(iii) Both (i) and (ii)
(iv) None of the above

(g) If $f : G \rightarrow G'$ is a homomorphism, then which of the following it true?

(i) $f(e) = e$
(ii) $f(e) = e'$
(iii) $f(e) >= \Phi$
(iv) $f(e) = 1$

(h) For which of the following does there exist a tree satisfying the specified constraints?

(i) A full binary tree with 31 leaves, each leaf of height 5
(ii) A rooted tree of height 3 where every vertex has atmost 3 children and there are 41 total vertices
(iii) A full binary tree with 11 vertices and height 6
(iv) A binary tree with 2 leaves and height 100

(i) For which of the following does there exist a graph *G*(V, *E*) satisfying the specified conditions?

(i) A tree with 9 vertices and the sum of the degrees of all the vertices is 18
(ii) A graph with 5 components, 12 vertices and 7 edges
(iii) A graph with 5 components, 30 vertices and 24 edges
(iv) A graph with 9 vertices, 9 edges and no cycles

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(j) The number of simple digraphs with |V| = 3 is

(i) 2^9 (iii) 2^7

(ii) 2^8 (iv) 2^6

Q.2 Solve both questions : **[6+8=14]**

(a) Let $f(x) = ax^2 + b$ and $g(x) = cx^2 + d$, where a, b, c and d are constants. Determine for which constants a, b, c and d the following equation holds: $f \circ g = g \circ f$

(b) Show that the relation $(x, y)\ R\ (a, b)$ such that $x^2 + y^2 = a^2 + b^2$ is an equivalence relation on the plane and describe the equivalence classes.

Q.3 Solve both questions : **[7+7=14]**

(a) Let $(G, \cdot)$ be a group, where $\cdot$ is usual multiplication operation on G. Then show that for any $x, y \in G$ following equations hold:

(i) $(x^{-1})^{-1} = x$

(ii) $(xy)^{-1} = y^{-1}x^{-1}$

(b) Construct the truth table for $[(p \vee q) \wedge (p \rightarrow r) \wedge (q \rightarrow r)] \rightarrow r$. Also show that above statement is a tautology by developing a series of logical equivalences.

Q.4 Solve both questions : **[6+8=14]**

(a) If A ={1, 2, 4}, B ={2, 5, 7} and C ={1, 3, 7}, find $(A \times B) \cup (A \times C)$.

(b) List the ordered pairs in the relation R from A ={1, 2, 3, 4} to B ={2, 3, 4, 5}, where $(a, b) \in R$ if and only if-

(i) $a = b$

(ii) $a + b = 5$

Q.5 Solve both questions : **[7+7=14]**

(a) Show that the set of integers with the composition $\circ$ and * defined by $a \circ b = a + b + 1$ and $a * b = ab + a + b$ is a ring.

(b) State and prove Lagrange's theorem.

Q.6 Solve both questions : **[7+7=14]**

(a) Define a relation R on the set Z of all integers as follows: m R n $\Leftrightarrow$ m+n is even for all m, $n \in Z$. Is R a partial order relation? Prove or give a counter example.

(b) Show that the group

(i) {1, 2, 3, 4}, X_5

(ii) {1, 2, 3, 4, 5, 6}, X_7

is cyclic.

Q.7 Solve both questions : **[7+7=14]**

(a) Let A ={0, 1, 2, 3}, R ={$(x, y) : x+y=3$}, S ={$(x, y) : 3|(x+y)$}, T ={(x, y) :max$(x,y)=3$}. Compute (i) $R \circ T$, (ii) $T \circ R$ and (iii) $S \circ S$.

(b) In a group of 70 cars tested by a garage in Delhi, 15 had faulty tyres, 20 had faulty breaks and 18 exceeded the allowable emission limits. Also, 5 cars had faulty tyres and brakes, 6 failed on tyres and emission, 10 failed on brakes and emissions, and 4 cars were unsatisfactory in all three respects. How many cars had no faults in these three checks? Draw an appropriate Venn diagram.

Q.8 Solve both questions : **[7+7=14]**

(a) Define the vertex connectivity and edge connectivity of a graph. Prove that for a graph G with n vertices and e edges, vertex connectivity $\le$ edge connectivity $\le 2e/n$.

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(b) Define the adjacency matrix of a graph. Find the rank of the regular graph with n vertices and with degree p ($< n$) of every vertex.

Q.9 Write short notes on any three of the following: **[14]**

(a) Multigraphs

(b) Planar graphs

(c) Cosets

(d) Ring polynomials

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